Lecture 37

Local equations and critical points

Uses.

Summary 8: Scalar product, quotient, chain and inverse derivative rules hold on their stated domains.
Summary 9: Fermat's theorem gives zero derivative at an interior differentiable extremum; the mean value theorem controls finite increments.
Summary 11: A scalar C^2 function has a degree-1 Taylor formula with a Lagrange second-derivative remainder.
Summary 12: Euclidean norms and Cauchy–Schwarz bound finite coordinate sums.
Summary 14: Polar coordinates satisfy $u = r\cos t$ and $v = r\sin t$.
Summary 18: Rank is the number of pivot columns. Summary 20 shows that a square matrix is invertible exactly when every column is a pivot column.
Summary 21: A square matrix is invertible exactly when its determinant is nonzero. Together with the preceding summaries, this is equivalent to full rank.
Summary 36: Differentiability gives a Jacobian linear approximation. C^1 implies differentiability, and Jacobians satisfy the matrix chain rule. Scalar gradients give directional derivatives.

Objectives. *A nonsingular derivative allows local equations to determine unknown coordinates. We state the precise existence theorems used for that step. Second derivatives then describe how a scalar function differs from its linear approximation. Their quadratic term classifies nondegenerate critical points in two variables.*

By the end of this lecture you will be able to:

- Apply the cited inverse and implicit function theorems with their open-domain, C^1 and nonsingularity hypotheses.
- Prove equality of mixed second partial derivatives for a C^2 scalar function.
- Prove the quadratic Taylor formula with a full norm remainder.
- Classify a planar critical point by the Hessian determinant when it is nonzero, and retain the inconclusive zero-determinant case.

37.1 Local inversion

The inverse of a linear map exists when its matrix is nonsingular. For a nonlinear map, an invertible derivative gives an inverse in a neighbourhood, rather than necessarily on the whole domain. A *neighbourhood* here means an open set containing the specified point.

Theorem 37.1 (Inverse function theorem, cited). *Let $U \subset \mathbb{R}^n$ be open and let $F : U \to \mathbb{R}^n$ be C^1. If $a \in U$ and $\det J_F(a) \neq 0$, there are neighbourhoods $V \subset U$ of a and W of $F(a)$ for which $F : V \to W$ is a bijection with a C^1 inverse $H : W \to V$. The neighbourhoods can be chosen so that*

$$J_H(F(x)) = J_F(x)^{-1} \qquad (x \in V).$$

The local existence assertion is cited [C].

The source is MIT 18.024, Chapter C notes, p. C12. The matrix formula is also forced by the proved chain rule: $H(F(x)) = x$ gives $J_H(F(x))J_F(x) = I$. This identity uses matrix inversion, not separate reciprocals of matrix entries. The cited theorem provides the local inverse whose existence is needed before differentiating.

Example 37.2 (Local coordinates with a global periodic ambiguity). Consider $F(s,t) = (e^s \cos t, e^s \sin t)$ on $\mathbb{R}^2$. Its Jacobian and determinant are

$$J_F(s,t) = \begin{pmatrix} e^s \cos t & -e^s \sin t \\ e^s \sin t & e^s \cos t \end{pmatrix}, \qquad \det J_F(s,t) = e^{2s} > 0.$$

Near $(0,0)$ an explicit inverse, for $u > 0$, is

$$H(u,v) = \left(\frac{1}{2} \ln(u^2 + v^2),\ \arctan \frac{v}{u} \right).$$

Indeed, restrict t to $(-\pi/2, \pi/2)$, where cosine is positive. The radius of $(u,v) = F(s,t)$ is e^s and its angle in that interval is $\arctan(v/u) = t$. Conversely those radius and angle formulas give $F(H(u,v)) = (u,v)$ for $u > 0$. This verifies a C^1 inverse on the strip and right half-plane directly. At $(0,0)$ the forward Jacobian is I, so the inverse Jacobian at $(1,0)$ is also I. Despite the nonzero determinant everywhere, the full map is not injective: $F(s, t + 2\pi) = F(s,t)$. The theorem's local conclusion cannot be replaced by a global one.

37.2 Implicit equations

Let G be a C^1 map from an open subset of $\mathbb{R}^n \times \mathbb{R}^m$ to $\mathbb{R}^m$. Write its input as (x,y), with $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. The matrices $D_x G$ and $D_y G$ are the blocks of its Jacobian containing partial derivatives in the x and y coordinates, respectively. Their sizes are $m \times n$ and $m \times m$. The *implicit function problem* asks whether $G(x,y) = 0$ determines y as a function of x near one specified solution.

Theorem 37.3 (Implicit function theorem, cited). *Suppose G is C^1 on its open domain, (a,b) belongs to that domain, $G(a,b) = 0$, and the square matrix $D_y G(a,b)$ is invertible. There are neighbourhoods V of a and W of b, with $V \times W$ in the domain, and a unique C^1 map $\varphi : V \to W$ such that*

$$G(x,y) = 0 \iff y = \varphi(x) \qquad ((x,y) \in V \times W).$$

After choosing sufficiently small neighbourhoods, its derivative is

$$J_\varphi(x) = -D_y G(x, \varphi(x))^{-1} D_x G(x, \varphi(x)).$$

The local existence assertion is cited [C].

The source is the MIT 18.022 handout *The implicit function theorem*, pp. 1–2. It states the same result for the level $G(a,b)$; here that level is zero. The derivative identity follows from the matrix chain rule on $G(x,\varphi(x))=0$: $D_xG+D_yG\,J_\varphi=0$ at $(x,\varphi(x))$. Continuity of the Jacobian and of its determinant permits shrinking so that the inverted block stays nonsingular. The neighbourhood W restricts the dependent coordinates; the theorem does not assert uniqueness among all possible remote solution branches.

Example 37.4 (A graph given by a logarithmic equation)**.** Consider $G(x,y)=y+\ln y-x$ on the open domain $y>0$. At $(a,b)=(1,1)$ its value is zero and $G_y=1+1/y=2\neq 0$. The theorem therefore supplies a unique local graph $y=\varphi(x)$ through $(1,1)$ [C]. The chain-rule formula gives

$$\varphi'(x)=-\frac{-1}{1+1/\varphi(x)}=\frac{\varphi(x)}{1+\varphi(x)},\qquad \varphi'(1)=\frac{1}{2}.$$

The derivative is known without an explicit formula for the graph. The formula and the graph remain on the positive-y domain where the logarithm is defined.

37.3 Mixed second derivatives

A scalar function is C^2 on an open set when all its partial derivatives through order two exist and are continuous there. Write $\partial_j\partial_i f$ for differentiation first in coordinate i and then in coordinate j. Its *Hessian* is the matrix $H_f=(\partial_j\partial_i f)_{1\le i,j\le n}$, the Jacobian of its gradient treated as a column. We prove that this matrix is symmetric under the C^2 hypothesis.

Theorem 37.5 (Schwarz's theorem)**.** *If f is C^2 on an open set $U\subset\mathbb{R}^n$, then $\partial_j\partial_i f(a)=\partial_i\partial_j f(a)$ for all $a\in U$ and all indices i,j.*

Proof. There is nothing to prove for equal indices. For distinct indices, fix the other coordinates and work in a sufficiently small coordinate rectangle inside U. Write the varying coordinates as (x,y) and the fixed point as (a,b). For nonzero small increments h,k, form the rectangular difference

$$\Delta=f(a+h,b+k)-f(a+h,b)-f(a,b+k)+f(a,b).$$

Apply the scalar mean value theorem in x to $f(x,b+k)-f(x,b)$. Then apply it in y to the resulting difference of x partial derivatives. The two applications give

$$\frac{\Delta}{hk}=\partial_y\partial_x f(a+\theta h,b+\eta k)$$

for some $0<\theta,\eta<1$. Applying the theorem in the opposite order gives $\Delta/(hk)=\partial_x\partial_y f(a+\widetilde{\theta}h,b+\widetilde{\eta}k)$, with possibly different intermediate points. As $(h,k)\to(0,0)$ through nonzero increments, both intermediate points tend to (a,b). Continuity of the two mixed partials makes their limits equal. This proves equality at the fixed point, and the point and coordinate pair were arbitrary. ∎

The intermediate points need not agree. Their convergence to the same point, together with continuity of the second partials, is the step that justifies exchanging their order.

37.4 The quadratic Taylor approximation

Theorem 37.6 (Second-order Taylor formula)**.** *Let $f : U \to \mathbb{R}$ be C^2 on an open set and let $a \in U$. For column increments $h \in \mathbb{R}^n$ tending to zero,*

$$f(a+h) = f(a) + \nabla f(a) \cdot h + \frac{1}{2} h^{\mathsf{T}} H_f(a) h + o(\|h\|^2).$$

The remainder divided in absolute value by $\|h\|^2$ tends to zero as $h \to 0$, uniformly over all approaches.

Proof. For sufficiently small h, the segment $a + th$, $0 \le t \le 1$, lies in U. Set $g(t) = f(a+th)$. The chain rule, first for f and then for its gradient, gives

$$g'(t) = \nabla f(a+th) \cdot h, \qquad g''(t) = h^{\mathsf{T}} H_f(a+th) h.$$

The scalar degree-1 Taylor formula on this segment gives, for some $0 < \theta < 1$, $g(1) = g(0) + g'(0) + g''(\theta)/2$. Subtract the proposed quadratic term. With $E(h) = H_f(a+\theta h) - H_f(a)$, the remaining error is $h^{\mathsf{T}} E(h) h/2$. The maximum $\varepsilon(h)$ of the absolute values of the finitely many entries of $E(h)$ tends to zero by continuity, since $\|\theta h\| \le \|h\|$. Consequently

$$|h^{\mathsf{T}} E(h) h| \le \varepsilon(h) \left(\sum_i |h_i| \right)^2 \le n\varepsilon(h) \, \|h\|^2 .$$

This proves the full remainder estimate. At $h = 0$ the equality holds with zero remainder. ∎

For example, the first coordinate of the earlier map is $f(x,y) = e^x \cos y$. At the origin its value, gradient and Hessian are

$$f(0) = 1, \qquad \nabla f(0) = (1, 0), \qquad H_f(0) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $f(x,y) = 1 + x + (x^2 - y^2)/2 + o(x^2 + y^2)$ near zero. The nonzero linear term is retained; a Hessian alone does not describe all changes at a point which is not critical.

37.5 Critical points in two variables

A point a of a differentiable scalar function is *critical* when $\nabla f(a) = 0$. A *local minimum* means that $f(x) \ge f(a)$ for all x in some ball about a; it is *strict* when the inequality is strict for $x \ne a$. Maxima reverse these inequalities. At an interior local extremum, each coordinate restriction has a scalar extremum. Fermat's theorem makes every partial derivative zero, so the point is critical. A critical point is a *saddle point* when every ball about it contains points whose values are above $f(a)$ and points whose values are below $f(a)$.

At a planar critical point, write the symmetric Hessian as $H = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$ and let $\Delta = AC - B^2$. The quadratic expression is $q(u,v) = Au^2 + 2Buv + Cv^2$. When $A \ne 0$, completing the square gives

$$q(u,v) = A\left(u + \frac{B}{A} v\right)^2 + \frac{\Delta}{A} v^2.$$

Theorem 37.7 (Planar Hessian test)**.** *Let f be C^2 near a planar critical point a. If $\Delta > 0$ and $A > 0$, then a is a strict local minimum. If $\Delta > 0$ and $A < 0$, it is a strict local maximum. If $\Delta < 0$, it is a saddle point. If $\Delta = 0$, this test makes no classification.*

Proof. Suppose $A > 0$ and $\Delta > 0$. Put $b = B/A$ and $d = \Delta/A > 0$. The completed square has a uniform positive lower bound. Indeed,

$$u^2 + v^2 \le 2(u + bv)^2 + (2b^2 + 1)v^2 \le M\big((u + bv)^2 + v^2\big), \qquad M = \max(2, 2b^2 + 1).$$

Thus $q(u, v) \ge c(u^2 + v^2)$, where $c = \min(A, d)/M > 0$. The Taylor formula at the critical point gives $f(a + h) - f(a) = q(h)/2 + r(h)$ with $|r(h)| \le c \|h\|^2 /4$ for all sufficiently small h. The difference is then at least $c \|h\|^2 /4 > 0$ for nonzero h, proving a strict local minimum. Applying this argument to $-f$ proves the maximum case when $A < 0$ and $\Delta > 0$.

If $\Delta < 0$ and $A \neq 0$, the directions $(1, 0)$ and $(-B/A, 1)$ give quadratic values A and Δ/A, which have opposite signs. If $A = 0$, then $B \neq 0$. Setting $v = 1$ and taking $u = (1 - C)/(2B)$ or $u = (-1 - C)/(2B)$ gives quadratic values 1 and -1. In either case choose fixed directions with opposite nonzero values of q. Along each direction $h = tw$, Taylor's remainder is $o(t^2)$, so its absolute value is eventually less than $|q(w)|t^2/4$. The sign of $f(a + tw) - f(a)$ is therefore the sign of $q(w)$ for sufficiently small nonzero t. Both signs occur arbitrarily near a, proving the saddle conclusion.

For $\Delta = 0$, the functions $x^4 + y^4$, $-x^4 - y^4$ and $x^4 - y^4$ all have zero gradient and zero Hessian at the origin. They have, respectively, a strict minimum, strict maximum and saddle there. Their signs on punctured neighbourhoods verify these three conclusions directly. Thus the zero-determinant data do not determine a classification. ∎

Figure 37.1 shows level curves of the polynomial in the following example.

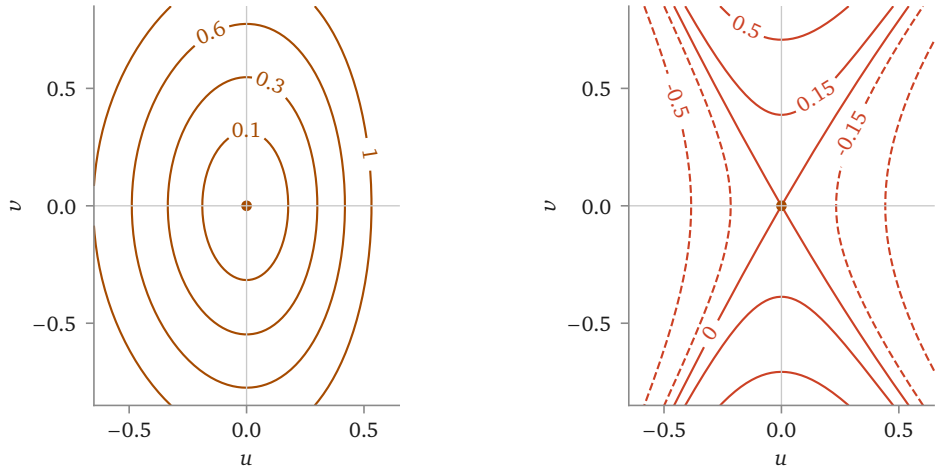


Figure 37.1: Level curves near the critical points of the worked cubic function. The left panel uses $u = x - 1$, $v = y - 2$ near the minimum; the right uses $u = x + 1$, $v = y - 2$ near the saddle. Labels give the change from the critical value; negative levels are dashed.

Example 37.8 (A local minimum and a saddle)**.** Find and classify the critical points of $f(x, y) = x^3 - 3x + y^2 - 4y$. The gradient equations are $3x^2 - 3 = 0$ and $2y - 4 = 0$, so the only critical points are $(1, 2)$ and $(-1, 2)$. Their Hessians and values are

$$H_f(1, 2) = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}, \quad f(1, 2) = -6, \qquad H_f(-1, 2) = \begin{pmatrix} -6 & 0 \\ 0 & 2 \end{pmatrix}, \quad f(-1, 2) = -2.$$

The first determinant is 12 with positive first diagonal entry, giving a strict local minimum. The second determinant is -12, giving a saddle. The local minimum is not a global minimum: with $y = 2$ fixed, the cubic term makes $f(x, 2) \to -\infty$ as $x \to -\infty$.

Summary 37.

- The inverse theorem requires a C^1 map on an open domain and a nonsingular square Jacobian. It supplies a local C^1 inverse [C]. The implicit theorem requires a C^1 map on an open domain, a specified solution and an invertible dependent-variable block, and supplies a unique local graph [C].
- Theorem 37.5: continuous second partial derivatives commute, so a C^2 scalar function has a symmetric Hessian.
- Theorem 37.6: the quadratic approximation has remainder $o(\|h\|^2)$ as $h \to 0$, uniformly over all approaches. It is proved from a scalar Taylor formula on the line segment.
- Theorem 37.7: at a planar C^2 critical point, positive Hessian determinant gives a strict minimum or maximum according to the first diagonal sign. A negative determinant gives a saddle; zero is inconclusive.

Sheet 37 develops local coordinate calculations and constrained extrema. The constrained existence arguments retain the implicit theorem's cited status. Lecture 38 uses Jacobians to change coordinates in double integrals.

References

MIT OpenCourseWare, *18.024 Multivariable Calculus with Theory*, Spring 2011, Chapter C notes, p. C12 (inverse theorem [C]), course notes. MIT *18.022 Multivariable Calculus*, *The implicit function theorem*, pp. 1–2 (implicit theorem [C]), course handout. J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §14.7 (critical points).